

The ethics of generative AI in software engineering

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1 Introduction

Software engineering is a field that has developed most of today's infrastructure in some way or another, be it via the implementation of a medical device to the websites that allow us to connect the world. However, these projects need to manage several stakeholders, engineers and budgets to make things come to reality. With the improvements in generative AI (GenAI for short), there has been a shift towards trying to incorporate this new tool into the software development pipeline. This tool has been adopted by industry leaders such as Google, Microsoft, Amazon and more. [17, 26, 28].

Given the workings of generative AI, there are several key concerns: The displacement of people for compute centers, the correctness of the code

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generated and the damage to growth spaces for future engineers. This essay will try to answer the question:

“Is it ethical for software engineers to use modern generative AI applications to generate code that is used, in whole or in part, for commercial applications?”

2 Core Issues

As discussed in the introduction, there are several concerns with the use of generative AI in software development:

2.1 Code correctness

One of the major issues that stems from using GenAI is the fact that GenAI doesn't *reason* like humans do given its answer is generated via approximating using other good answers in the source. Programming is a task that requires discrete reasoning as the steps will directly relate to the next one in the solution [6]. This causes several issues when it comes down to the reliability of the software being written with the help of GenAI [15].

According to a 2024 study conducted by the International University of Applied Sciences, the amount of code generated is about 97% of the times executable. While this is a high rate, the issues start to stem from the correctness, where about 21% of the generated lines of code were fully correct and about 18% were able to be corrected minimally to achieve the desired result [15]. While the methods of the study focused on smaller programs with simpler functionality, the risks of failure can be

extrapolated to larger systems where the codebase is larger and more side effects need to be taken into account when submitting changes.

A production instance of code correctness not being up to par is Amazon's outages in 2026. The outage of major Amazon services was reported to be caused by a GenAI tool used for code. Repeated instances of this are being prevented by strengthening the guard rails present for change submissions, specially those that used GenAI in their development. However, it is important to highlight major services like Amazon Web Services (AWS) went down due to GenAI changes [21]. The issue here is not AWS being down but instead the clients of the service being collateral damage of such infrastructure failure without a clear accountability chain.

As seen with AWS, substantial damage can be done to infrastructure even when there are senior engineers in the production process. The risks become more dangerous with issues like GenAI hallucination [14] or the use of GenAI code in devices and systems that require critical safety such as biotechnology, embedded systems in transportation and more. In those cases, a model converging to the wrong answer can cost lives and without a clear accountability chain, it is difficult to enforce justice and repair damages. There are several issues with deciding how to hold accountability as mentioned in an article by IBM: "With the C-suite? With the whole team? If you have accountability that's spread over the entire organization, everyone can't end up in jail. Ultimately, shared accountability often leads to no accountability." [9].

2.2 Damage to growth spaces for entry level engineers

Another major issue with GenAI for software engineering is the fact that they reduce the spaces where less experienced engineers build the knowledge to move forward. As stated before, GenAI doesn't reason and this is a similar issue with less experienced engineers, however, the key difference is that human engineers can improve with time and build up the discrete reasoning which

allows them to construct correct programs that they haven't seen before. [22]

As mentioned in an article by computer world, it is mentioned how Amazon cut around 14000 jobs because of the developments in GenAI technology allowing them to replace the humans with machines [26]. This presents a major issue as there is still a need for engineers in the production pipeline, even more so when mistakes like the outages necessitating more human reviews of the submitted code [21]. This pattern is industry wide as there is a 25% decrease in entry level tech job hiring and a decrease of about 30% in internship positions [22].

In an blog post by Stack Overflow, the author discusses the importance of hiring entry level engineers as this facet of the tech industry is apprenticeship heavy given the complex systems that are being produced and maintained that are simply not available in an academic context. The article also concludes with the call to action to senior engineers and managers highlighting how there will be no other way to train entry level engineers beyond hiring them [18].

2.3 Displacement of people

This is a more general issue with the usage of GenAI but applies heavily to expensive tasks like generating the code for a complex program. The infrastructure that GenAI needs is costly in several resources [20, 29]:

- About 22% of all electricity used by US households per year.
- "Chan School of Public Health found that the carbon intensity of electricity used by data centers was 48% higher than the US average."
- "Large data centers can consume up to 5 million gallons of water per day, equivalent to the water use of a town populated by 10,000 to 50,000 people."

In addition to the resource costs, data centers require space to be built. In the past year, there have been about 831 data centers built across the world to meet the GenAI demand. These sites have been constructed across the world [7]. While this

will decrease resources available for communities where the centers are built, the data centers will affect communities in developing countries the most. This is due to companies concealing their identities to purchase the land, and in some cases, not needing to disclose the resources they will be consuming, such as Microsoft's license to exploit a well that has been overexploited in Mexico [5].

3 Stakeholders

3.1 General Population

3.2 Software engineers

3.3 Companies that use generative AI

3.4 Companies that make generative AI tools

4 Current state of generative AI

With understanding of the costs of this technology it is also important to acknowledge its usage. By understanding the applications of the technology, it will be possible to weigh out the benefits and disadvantages of spending resources on this technology.

4.1 Usage in research

There's been a shift in academia to use more GenAI to speed up different elements of the research pipeline [1]. A key project that will be mentioned in this essay is the usage to create the implementation for the ZOZO's (a Japanese clothing company) cloth physics solver [2, 3].

4.2 Usage in open source

In open source projects there is a current issue where the lead contributors are being swarmed by pull requests with low quality AI generated code [16]. This puts a strain in projects' ability to survive as the leads are often people who volunteer

and don't have the time to go through all the low quality code to pick out and accept contributions to the project.

4.3 Usage in private companies

Across different companies, there are several philosophies, some companies are encouraging their developers to use GenAI and needing GenAI to strengthen the code review process [8] and other companies avoid using GenAI as they believe it damages the quality of their products [13].

5 Technical Background

5.1 Machine learning

Machine learning is a subset of artificial intelligence that uses statistical inference to adapt responses based on previous and new knowledge. This is a very powerful type of program as it provides adaptability at no code architecture change, however, these systems improve drastically the more data it has available and analyzed [10].

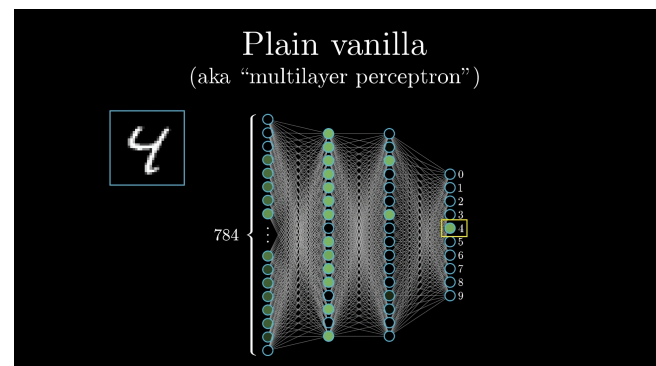


Figure 1: Neural network made to recognize numbers 0 to 9 on a 28 by 28 grid [23].

Models work via the implementation of neural networks. A neural network is composed of *neurons*, which are simply just a number that represents the presence of a component in the data (the circles in the image 1). A vector of neurons represents a layer, this is used to represent different

components of the data. Neural networks are normally composed of several layers each neuron connected to another via weights (the lines in image 1). This allows the model to encode the activation of a neuron based on a linear combination of the weights and neurons of the previous layer. In the image 1 the activation is represented by how green each circle is. Each layer encodes different sets of properties about the data, in the case of figure 1 the first layer represents how bright each pixel is while the next one represents relationships in the distributions. The first layer is one of the few layers that can be explicitly linked to the data as the activation is entirely determined by the model maker. This is because these linear combinations can be fine tuned with a bias term to ensure that the model produces better results [23].

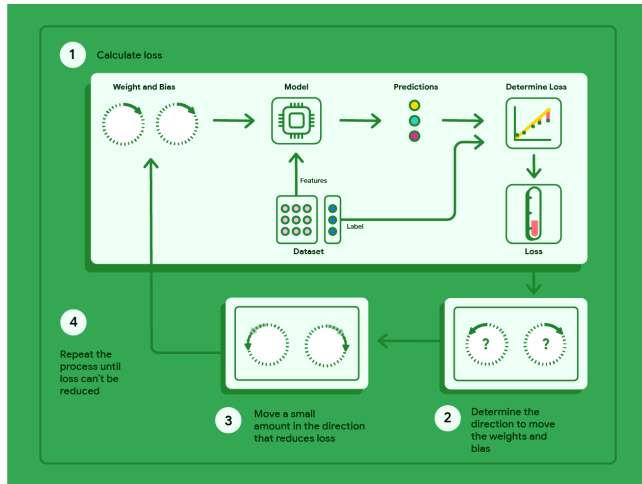


Figure 2: The model training pipeline [11]

The training process for a model involves utilizing data that is labelled to be the right response and iteratively improving all of the model's weights and biases (denoted by $\vec{w}$). This process of improvement utilizes cost function and gradient descent. The cost function (ϵ) describes the error of the current model based on the given parameters, this error is normally averaged to improve across all input types instead of only one [11]. Gradient descent is an iterative optimization method defined as such:

$$\vec{w}' = \vec{w} + \gamma \begin{bmatrix} \frac{\partial \epsilon}{\partial w_1} \\ \vdots \\ \frac{\partial \epsilon}{\partial w_n} \end{bmatrix}$$

This method is used for steps 2 and 3 in figure 2 as it will calculate the direction to move the weights and biases and move there. This process is repeated until the change in error between iterations is less than desired [11]. While this only adjusts the weights, there still needs to be a process of back-propagation. This involves finding which parameters affect the output the most and then adjusting the values to cover the edge cases left after the gradient descent is done ¹ [24].

5.2 Large Language Models

Large Language Models (LLMs) are models that are trained on natural and computer language in order to be able to generate an output based on verbal (text and/or audio) input. These models use the relationships encoded with machine learning to output next most likely word [25].

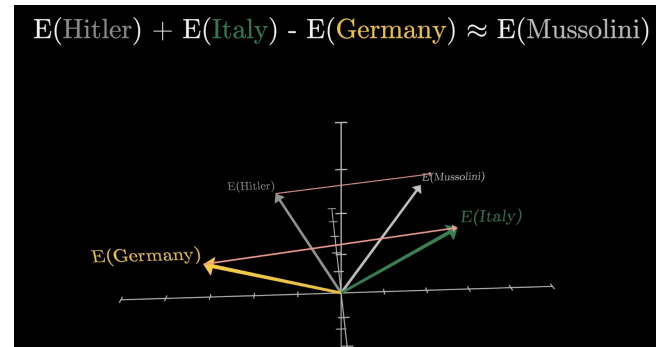


Figure 3: Visuals of how the embedding matrices transform different words into different "word spaces" [25]

When the user submits a prompt, the model will turn the prompt into tokens. Tokens are small chunks of the prompt that can be turned into vector values (each word being like a pixel in figure 1).

¹The math of this algorithm is more complex therefore it will not be explored in depth for this essay

This is done with an embedding matrix, which describes the conversions between a token and their respective vector value. This vector value, like discussed in the machine learning section, represents a set of relationships be it about the word itself or the semantic context of the word [25].

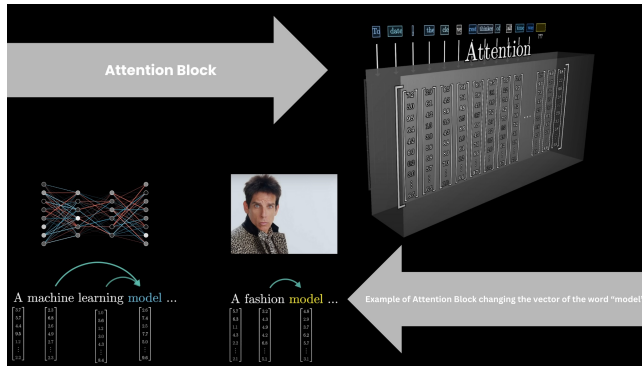


Figure 4: Visuals of how the attention block uses and represents context numerically [25]

After that, each vector will affect each other to ensure that the context is taken into account, this is known as the attention block (see figure 4) [25].

After finalizing the contextualized vectors, the model will then pass each vector separately through the layers of the neural network. This will result with a final vector like the one in figure 1, but the vector will represent a scalar of how relevant the word is.

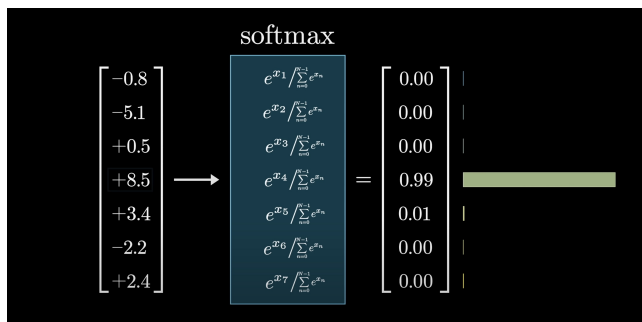


Figure 5: How the scalar values are turned into a probability distribution [25]

As the scalars are not a probability, the procedure in figure 5 is performed. This then allows the model to pick either the most likely word with a bit of

added randomization to mimic natural language [25].

This process is conducted iteratively until the response is fully formed. The nice thing about this model is how it can turn the set of input tokens into a set of different parameters used to construct an output. While this example focused on generating text, this extends to code with the main difference being that the output vector is not composed of words or punctuation but symbols and language tokens (such as brackets or keywords) [27].

5.3 Architecture of GenAI coding tools

When utilizing generative AI in the context of a software project, there are additional considerations that must be met to meet the desired output. To better understand this fine tuning, this essay will focus on Claude Code's approach as a case study.

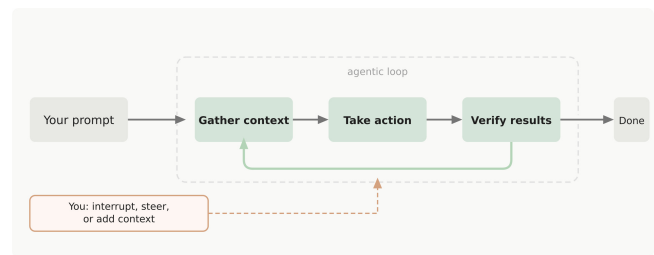


Figure 6: Claude Code's prompt processing architecture [4]

When a developer prompts Claude Code, the tool will start by gathering context of the problem. This involves analyzing the codebase for relevant symbols and classes that are needed to complete the required prompt. Then the tool will execute actions such as generating text or code. Lastly, because Claude has permissions to interact with the terminal, it will try to verify the results by either compiling the code, running the code and verifying any output produced by the program [4].

While not all developers directly allow GenAI tools to access their codebase like in figure 6, this is the most effective at producing responses as it

is able to gather more information and test the generated content.

6 Ethical Analysis

Given the broadness of the issue presented in the essay, this section will be split into the core issues presented in section of the same name.

6.1 Software Reliability

As discussed in the technical section, AI utilizes the relationship between the provided prompt and the relationships it can build with the data to generate the code snippet. There is a major issue in this generation process, it doesn't perform logical reasoning in order to ensure that the code is correct. Tools like Claude utilize the added knowledge of the database, compiler errors and program output to ensure the program performs what is desired. However, this poses a critical issue, there is no guarantee that the side effects that stem from the lines being executed (such as memory allocation, pointer dereferencing, etc) don't harm the project or the users. This can be seen in several cases where the models are not trained with code of the quality desired, which in turn leads to bugs [12].

According to SOURCE HERE, companies are shifting to GenAI to reduce costs. Under Kantianism, this is ignoring other people's humanity in multiple ways. First, the open source projects that were used to train the model were not asked for their permission (REPHRASE). Second, it disregards the user's humanity by prioritizing the company's profits over the correctness of the product they sold to the customer.

In the lens of act utilitarianism, there are also major concerns. If the goal is to maximize utility, creating infrastructure that fails to consistently deliver can reduce the overall benefits provided to the consumers. When the software written using GenAI is delivered to millions of people, the benefits that the engineers and corporate leads receive do not outweigh the downsides of the consumer base. In another framing, if the benefits were to be

mainly monetary, the damages to society end up costing more in the long term, be it through the increase in electricity [20] or the damage caused by failures in the delivered products [19]. In the latter case, the accountability chain is also not clearly defined as to who is responsible or why the error happened as it could be a GenAI hallucination or user error in the prompt. This can cause major issues when trying to resolve the damages caused to the users and other individuals.

From these two frameworks it can be seen that there are major issues that must be solved before GenAI can provide benefits to developers and companies that will outweigh the costs to the consumer. However, in the current state it fails both at Kantianism and Act Utilitarianism. If the issues were addressed, it would maybe be able to handle the issues presented by Act Utilitarianism.

6.2

7 Conclusion

8 Beyond the superficial

9 Final remarks

Do not miss sections entirely, fix the grammar mistakes, and ensure that the sources are pulled out properly.

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